

Password Strength Analyzer

Project Overview

Develop a tool that evaluates the strength of user-entered passwords.

Key Features

- Check password length, complexity, and uniqueness
- Suggest stronger password alternatives
- Optional database integration to prevent password reuse

Technologies Used

Python, Regular Expressions, String Processing, SQLite (Optional)

Folder Structure

```
password-strength-analyzer/  
├──  
│   ├── analyzer.py  
│   ├── passwords.db  
│   ├── requirements.txt  
│   └── README.md
```

Install Required Packages

```
pip install regex
```

analyzer.py

```
import re  
import random  
import string  
  
def check_password_strength(password):  
    score = 0  
    remarks = []  
  
    if len(password) >= 8:  
        score += 1  
    else:  
        remarks.append("Password should be at least 8 characters long")  
  
    if re.search(r"[A-Z]", password):  
        score += 1  
    else:  
        remarks.append("Add at least one uppercase letter")  
  
    if re.search(r"[a-z]", password):  
        score += 1  
    else:
```

```

        remarks.append("Add at least one lowercase letter")

    if re.search(r"[0-9]", password):
        score += 1
    else:
        remarks.append("Add at least one number")

    if re.search(r"[!@#$$%^&*(),.?\"':{}|<>]", password):
        score += 1
    else:
        remarks.append("Add at least one special character")

    if score == 5:
        strength = "Strong"
    elif score >= 3:
        strength = "Medium"
    else:
        strength = "Weak"

    return strength, remarks

def generate_strong_password():
    characters = string.ascii_letters + string.digits + "!@#$$%^&*"
    password = ''.join(random.choice(characters) for i in range(12))
    return password

password = input("Enter Password: ")

strength, remarks = check_password_strength(password)

print(f"\nPassword Strength: {strength}")

if remarks:
    print("\nSuggestions to Improve Password:")
    for remark in remarks:
        print("-", remark)

if strength != "Strong":
    print("\nSuggested Strong Password:")
    print(generate_strong_password())

```

Security Features

Feature	Description
Password Length Check	Ensures minimum secure password length
Complexity Check	Checks uppercase, lowercase, numbers, and symbols
Password Suggestions	Generates stronger password alternatives
Optional Database Support	Prevents reuse of old passwords

Expected Outcome

Learn about password security and basic cryptography concepts.

Conclusion

The Password Strength Analyzer project helps users understand password security principles. It evaluates password strength based on complexity rules and suggests stronger passwords to improve account security.